EduCAP™ Worksheet

Alabama Comprehensive Assessment Program

Bush Hills STEAM Academy
Full STEAM Ahead

ACAP Dummy Worksheet — ELA G6

Subject: ELA | Grade: 6 | DOK Level: 2 | Standard: 6.CL.2

Name: _____ Date: _____ Period: _____

Ø=ÜÖ Reading Passage

The deep ocean, a realm often described as dark and mysterious, is home to one of nature's most dazzling secrets: bioluminescence. This incredible phenomenon, the production of light by living organisms, transforms the inky blackness into a spectacular light show. While many people might associate natural light with fireflies flickering on a summer evening, a vast majority of bioluminescent creatures reside hidden beneath the ocean waves, from the shallow coastal waters to the crushing depths of the abyssal plain. Scientists estimate that up to 90% of deep-sea organisms exhibit some form of bioluminescence, making it far more common in the ocean than on land. This biological light isn't just for show; it serves vital purposes for survival in a challenging environment.

So, how do these organisms create their own light? Bioluminescence is essentially a chemical reaction. Most commonly, it involves a molecule called luciferin, which is the compound that actually produces the light, and an enzyme called luciferase, which acts as a catalyst to speed up the reaction. When luciferin reacts with oxygen in the presence of luciferase (and sometimes other cofactors like ATP), energy is released in the form of cold light. Unlike a conventional light bulb, which generates a lot of heat, bioluminescent light produces very little heat, making it an incredibly efficient process. Some organisms produce these chemicals themselves, while others host symbiotic bacteria that generate the light for them.

The reasons behind this glowing ability are as varied as the creatures themselves. For many, bioluminescence is a crucial tool for **evasion** and defense. Some squid and shrimp, for instance, release a cloud of glowing fluid, similar to an octopus's ink, to startle or distract predators and then make a quick escape. Others use flashes of light as a 'burglar alarm,' attracting a larger predator to attack the initial threat. In the perpetual darkness of the deep sea, light can also be a powerful lure. The infamous anglerfish uses a bioluminescent 'fishing rod' (a modified dorsal fin spine called an esca) to attract unsuspecting prey directly to its massive, tooth-filled mouth.

Communication is another key function. Certain species of fish use specific patterns of flashes or glows to recognize members of their own species, distinguish between males and females, or even to warn rivals away from territory. For example, some ostracods (tiny crustaceans) release a series of glowing dots into the water, creating a unique 'light language' for mating rituals. This precise signaling ensures that they attract the correct mate in an environment where sight is otherwise limited. The ability to produce light, therefore, is not a random evolutionary quirk but a finely tuned adaptation that helps these organisms thrive.

Studying bioluminescence offers scientists a window into the complex adaptations of marine life and even holds

potential for human innovation. Researchers are investigating how these biological light systems could be used in medical imaging, environmental monitoring, or even sustainable lighting technologies. The deep ocean, once considered a barren wasteland, continues to reveal its secrets, reminding us of the extraordinary diversity and ingenuity of life on Earth. The living lights of the ocean are a testament to evolution's creative power, illuminating pathways not only for the creatures that produce them but also for our understanding of the natural world.

1. Based on paragraphs 3 and 4, which statement best explains the primary *advantage* bioluminescence provides to marine organisms?

A) It allows them to generate heat in the cold deep sea.
B) It helps them perform essential survival tasks like defense, hunting, and communication.
C) It makes them more visible to human researchers studying the ocean.
D) It provides a way for them to absorb sunlight in dark environments.

2. Select ALL the ways marine organisms use bioluminescence for defense, as described in paragraph 3.

(Select ALL that apply)

%¡ A) To attract smaller predators to protect them from larger ones.
%¡ B) To create a glowing 'smoke screen' to escape from threats.
%¡ C) To act as a 'burglar alarm' by drawing a larger predator to an attacker.
%¡ D) To signal to other species that they are poisonous.

3. In paragraph 3, the word "evasion" most closely means:

A) Attraction
B) Discovery
C) Avoidance
D) Communication

4. Explain how the anglerfish uses bioluminescence to solve a critical problem in its environment. Use specific evidence from the passage in your answer.

' Text-Dependent Writing — Question 5

5. The passage describes several distinct ways marine organisms utilize bioluminescence for survival. Write a multi-paragraph informative essay explaining how this phenomenon serves as a versatile adaptation for different marine creatures. Use specific evidence from the text to support your explanation, ensuring you discuss at least three different uses.

Answer Key

ACAP Dummy Worksheet — ELA G6 | 6.CL.2

1. B — Option B best summarizes the various survival advantages (defense, hunting, communication) discussed in paragraphs 3 and 4. Option A is incorrect because bioluminescence produces 'cold light' (paragraph 2). Option C is a benefit to humans, not the organisms. Option D is incorrect as the deep sea lacks sunlight.

2. B,C — Paragraph 3 explicitly states, 'Some squid and shrimp, for instance, release a cloud of glowing fluid... to startle or distract predators and then make a quick escape,' which supports option B. It also states, 'Others use flashes of light as a 'burglar alarm,' attracting a larger predator to attack the initial threat,' which supports option C. Option A misrepresents the 'burglar alarm' use, and option D is not mentioned in the passage.

3. C — The sentence in paragraph 3 states, 'For many, bioluminescence is a crucial tool for **evasion** and defense. Some squid and shrimp, for instance, release a cloud of glowing fluid... to startle or distract predators and then make a quick escape.' The context of 'defense' and 'make a quick escape' indicates that 'evasion' means avoiding or escaping from a threat.

4. [Short Response]

Rubric:

2 points: Clearly explains the problem (darkness, finding prey) and solution (bioluminescent lure) using specific evidence from the passage.

1 point: Provides a partial explanation or lacks specific textual evidence for both the problem and solution.

0 points: Does not answer the question or provides irrelevant information.

Sample Response:

The anglerfish lives in the deep ocean, an environment where it is perpetually dark, making it very difficult to find prey. To solve this critical problem, the anglerfish uses a bioluminescent 'fishing rod' (an esca) to attract unsuspecting prey directly to its mouth. This glowing lure helps the anglerfish hunt effectively in its dark habitat, as described in paragraph 3.

5. [Text-Dependent Writing]

Rubric:

4 Points (Exemplary): Provides a thorough, well-organized, multi-paragraph informative essay that clearly explains at least three distinct uses of bioluminescence for survival. Demonstrates a deep understanding of the text by integrating multiple pieces of specific, relevant textual evidence with accurate citations. Analysis is insightful and fully supports the explanation.

3 Points (Proficient): Provides a clear, organized, multi-paragraph informative essay that explains at least two distinct uses of bioluminescence for survival. Integrates relevant textual evidence with citations to support the explanation. Analysis is generally clear, though may lack some depth.

2 Points (Developing): Provides a basic multi-paragraph response that attempts to explain uses of bioluminescence. May discuss only one use or provide limited details for multiple uses. Textual evidence may be general, insufficient, or lack citations. Organization may be unclear.

1 Point (Emerging): Provides a minimal or fragmented response with little to no clear explanation of bioluminescence uses. Textual evidence is absent or irrelevant. Lacks organization.

0 Points (No Score): Response is completely off-topic, merely copies text, or is blank.

Sample Response:

The deep ocean, a realm of perpetual darkness, is home to creatures that have developed extraordinary adaptations to survive, one of the most remarkable being bioluminescence. This ability to produce light serves marine organisms in numerous crucial ways, acting as a versatile tool for defense, hunting, and communication. Without this unique adaptation, many deep-sea species would struggle to thrive in their challenging environment.

One significant way marine creatures use bioluminescence is for defense and evasion from predators. As explained in paragraph 3, some organisms, like squid and shrimp, can release a glowing fluid to create a 'smoke screen' that startles or distracts an attacker, allowing them to escape. Other species employ a 'burglar alarm' strategy, flashing light to attract a larger predator to attack their original threat. These defensive maneuvers demonstrate how light can be used not just to reveal, but also to conceal and confuse, ensuring survival in a dangerous world.

Beyond defense, bioluminescence is also a powerful tool for hunting and attracting prey. The anglerfish, for example, has evolved a specialized 'fishing rod' called an esca, tipped with bioluminescent bacteria, which it dangles in the dark waters. This glowing lure attracts unsuspecting prey directly to its waiting mouth, providing a highly effective hunting

mechanism where sight is otherwise impossible (paragraph 3). This predatory use highlights how bioluminescence can turn the disadvantage of darkness into a strategic advantage.

Finally, communication is another vital function of bioluminescence, especially for species recognition and mating. In the vast, dark ocean, finding a mate or identifying members of one's own species can be incredibly difficult. Paragraph 4 describes how certain fish use specific patterns of flashes to recognize their kind, and how ostracods create unique 'light language' signals for mating rituals. This precise signaling ensures successful reproduction and social interactions, proving that light can serve as a complex form of language.

In conclusion, bioluminescence is far more than just a beautiful display; it is a multi-faceted adaptation that underpins the survival of countless marine organisms. From evading predators and luring prey to facilitating intricate communication, the living lights of the ocean showcase evolution's ingenious solutions to the challenges of life in the deep. Understanding these varied uses helps us appreciate the complexity and interconnectedness of marine ecosystems.